AGENDA

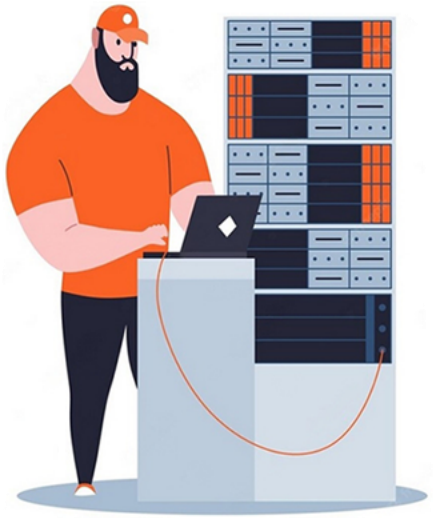
What is Name Resolver?

About DNS

About DNZ ZONE

About DNS SUB ZONE’s

About DNS QUERY



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WHAT IS NAME RESOLVER

WHAT IS NAME RESOLVER

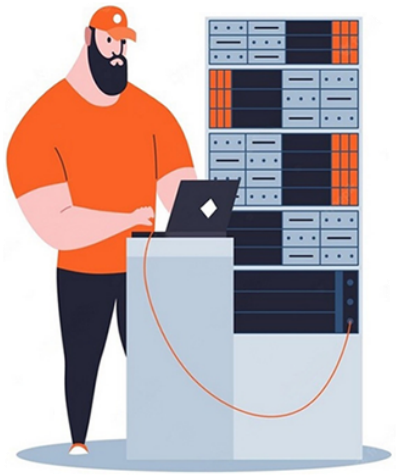


A name resolver is a system or service that helps translate human-readable names into machine-readable addresses or identifiers. It is commonly used in computing, particularly in networking and domain name systems (DNS).
Name Resolver is a technique by which you can resolve the name **IP to Name** or **Name to IP**.

- Name = CN (Computer Name)
- DN (Domain Name)
- FQDN (CN + DN)

There are three type of Name Resolvers following down: -

- BIND
- WINS
- DNS





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ABOUT DOMAIN NAME SERVICE

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BIND stands for **Berkeley Internet Name Domain**. The software was originally designed at the University of **California Berkeley** (UCB) in the early **1980s**. The latest version of BIND is BIND 9, first released in 2000. It was Unix based.

WINS stand for **Windows Internet Name Services**. WINS is developed by **Microsoft Corp**. It resolve the query by using **LMHost** file Name to IP. It is Windows operating system based.

DNS stands for **Domain Name System**. It is open standard service. In **2000** it was launched with Windows Server 2000. It resolve query Name to IP and IP to Name. It support all type of OS. DNS defines a hierarchical namespace, where each level of the namespace is separated by a "." DNS use query.



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ABOUT DNS ZONE

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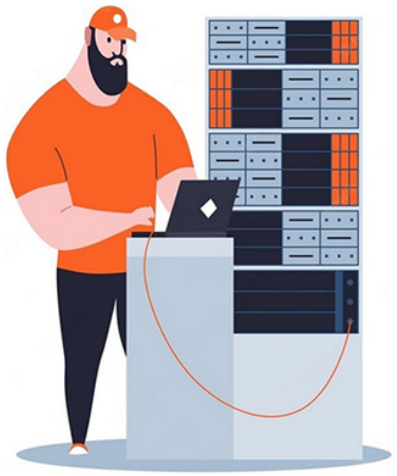


DNS work with Zone to resolve the query. There are two type of Zones.

- **Forward Lookup Zone:-** Contain host record, which contain host name to IP, addressing mapping information.
- **Reverse Lookup Zone:-** It contains mapping information about IPs to host.

These Zones work with **Sub-Zone**.

- **Primary Zone -** A Primary Zone (also called a Master Zone) is the authoritative source for a particular domain's DNS records. It contains the original, read-write copy of the zone data.
- **Secondary Zone -** A Secondary Zone is a read-only copy of a primary zone. It is used to provide redundancy and load balancing for DNS queries, ensuring that DNS resolution can continue even if the primary server is unavailable.
- **Stub Zone -** A Stub Zone is a type of DNS zone that contains only the essential records required to locate the authoritative DNS servers for another zone. It does not contain the complete set of DNS records for a domain like a primary or secondary zone, but only the NS (Name Server) records, the A (Address) records for those servers, and SOA (Start of Authority) record.
- **AD Integrated Zone** - An Active Directory Integrated Zone is a DNS zone that is stored and managed within Active Directory (AD) rather than being kept as a standalone file. This allows DNS data to be integrated with other AD data and benefit from AD's features, like security, replication, and fault tolerance.



ABOUT DOMAIN NAME SERVICE

DNS Records: These are the entries that store information about a domain, including its IP address. Common types of DNS records include:

- **A Record:** Maps a domain name to an IPv4 address.
- **AAAA Record:** Maps a domain name to an IPv6 address.
- **CNAME Record:** Maps one domain name to another.
- **MX Record:** Specifies mail servers for a domain.
- **PTR Record (pointer) used in RLZ:**
- **Alias Record (Nick Name of a host record):**
- **MX Record (used for mail server):**
- **SRV Record (used for chat server):**

